

torch.nn.utils.parametrizations.spectral_norm

https://pytorch.org/docs/stable/generated/torch.nn.utils.parametrizations.spectral_norm.html

Description:

Apply spectral normalization to a parameter in the given module. When applied on a vector, it simplifies to Spectral normalization stabilizes the training of discriminators (critics) in Generative Adversarial Networks (GANs) by reducing the Lipschitz constant of the model. σ is approximated performing one iteration of the power method every time the weight is accessed. If the dimension of the weight tensor is greater than 2, it is reshaped to 2D in power iteration method to get spectral norm. See Spectral Normalization for Generative Adversarial Networks .

Function Signature

```
torch.nn.utils.parametrizations.spectral_norm(module,  
name='weight', n_power_iterations=1, eps=1e-12, dim=None)  
[source]#
```

Parameters

Returns

Return type

Returns:

Module

Code Examples

```
>>> snm = spectral_norm(nn.Linear(20, 40))
>>> snm
ParametrizedLinear(
  in_features=20, out_features=40, bias=True
  (parametrizations): ModuleDict(
    (weight): ParametrizationList(
      (0): _SpectralNorm()
    )
  )
)
>>> torch.linalg.matrix_norm(snm.weight, 2)
tensor(1.0081, grad_fn=<AmaxBackward0>)
```

Notes & Warnings

NOTE This function is implemented using the parametrization functionality in `register_parametrization()`. It is a reimplementation of `torch.nn.utils.spectral_norm()`.

NOTE When this constraint is registered, the singular vectors associated to the largest singular value are estimated rather than sampled at random. These are then updated performing `n_power_iterations` of the power method whenever the tensor is accessed with the module on training mode.

NOTE If the `_SpectralNorm` module, i.e., `module.parametrization.weight[idx]`, is in training mode on removal, it will perform another power iteration. If you'd like to avoid this iteration, set the module to eval mode before its removal.

Detailed Documentation

Documentation

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Spectral normalization stabilizes the training of discriminators (critics) in Generative Adversarial Networks (GANs) by reducing the Lipschitz constant of the model. $\sigma(\sigma)$ is approximated performing one iteration of the power method every time the weight is accessed. If the dimension of the weight tensor is greater than 2, it is reshaped to 2D in power iteration method to get spectral norm.

See Spectral Normalization for Generative Adversarial Networks .

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If the `_SpectralNorm` module, i.e., `module.parametrization.weight[idx]`, is in training mode on removal, it will perform another power iteration. If you'd like to avoid this iteration, set the module to eval mode before its removal.

`module (nn.Module)` – containing module

`name (str, optional)` – name of weight parameter. Default: "weight".

`n_power_iterations (int, optional)` – number of power iterations to calculate spectral norm. Default: 1.

`eps (float, optional)` – epsilon for numerical stability in calculating norms. Default: 1e-12.

`dim (int, optional)` – dimension corresponding to number of outputs. Default: 0, except

for modules that are instances of `ConvTranspose{1,2,3}d`, when it is 1

The original module with a new parametrization registered to the specified weight